

## **TASK 4 : Responsible AI and Model Interpretation**

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Internship Platform: InternSpark (Powered by Alfido Tech)

### **1. Objective**

Analyze fairness, bias and explainability of the Resume Classification model using Responsible AI techniques.

### **2. Project Overview**

The trained model was analyzed using explainability techniques and dataset bias analysis. Category imbalance was identified and mitigation strategies were proposed.

### **3. Technologies Used**

Python  
TensorFlow  
Pandas  
NumPy  
Matplotlib  
LIME  
Jupyter Notebook

### **4. Files Used**

fairness\_analysis.ipynb  
README.md

### **5. GitHub Code Link**

Paste your GitHub repository link here after uploading the project.

### **6. Analysis Performed**

1. Generated LIME explanations
2. Checked dataset imbalance
3. Visualized category distribution
4. Proposed mitigation strategies

### **7. Results and Screenshots**

Dataset Preview

|    | Category     | Resume                                            |
|----|--------------|---------------------------------------------------|
| 0  | Data Science | Skills * Programming Languages: Python (pandas... |
| 1  | Data Science | Education Details \nMay 2013 to May 2017 B.E ...  |
| 2  | Data Science | Areas of Interest Deep Learning, Control Syste... |
| 3  | Data Science | Skills â R â Python â SAP HANA â Table... |
| 4  | Data Science | Education Details \n MCA YMCAUST, Faridabad...    |
| 5  | Data Science | SKILLS C Basics, IOT, Python, MATLAB, Data Sci... |
| 6  | Data Science | Skills â Python â Tableau â Data Visuali... |
| 7  | Data Science | Education Details \n B.Tech Rayat and Bahra ...   |
| 8  | Data Science | Personal Skills â Ability to quickly grasp t... |
| 9  | Data Science | Expertise â Data and Quantitative Analysis â... |
| 10 | HR           | TECHNICAL SKILLS â Typewriting â TORA â ... |
| 11 | HR           | I.T. Skills â Windows XP, Ms Office (Word, E... |
| 12 | HR           | Education Details \n BA mumbai UniversityHRS...   |
| 13 | HR           | Education Details \nJune 2012 to May 2015 B.A ... |
| 14 | HR           | Education Details \nJune 2012 to May 2015 B.A ... |
| 15 | HR           | Education Details \n BBA lovely professional...   |
| 16 | HR           | Education Details \n MBA ACN College of engi...   |
| 17 | HR           | KEY SKILLS: â Computerized accounting with t... |
| 18 | HR           | Training in Special Education (Certificate Cou... |

Explanation: Dataset Preview generated during fairness and explainability analysis.

### Category Labels

```
['Advocate' 'Arts' 'Automation Testing' 'Blockchain' 'Business Analyst'
 'Civil Engineer' 'Data Science' 'Database' 'DevOps Engineer'
 'DotNet Developer' 'ETL Developer' 'Electrical Engineering' 'HR' 'Hadoop'
 'Health and fitness' 'Java Developer' 'Mechanical Engineer'
 'Network Security Engineer' 'Operations Manager' 'PMO' 'Python Developer'
 'SAP Developer' 'Sales' 'Testing' 'Web Designing']
```

Explanation: Category Labels generated during fairness and explainability analysis.

### Model Summary

| Layer (type)                  | Output Shape    | Param # |
|-------------------------------|-----------------|---------|
| embedding (Embedding)         | (None, 400, 64) | 320,000 |
| bidirectional (Bidirectional) | (None, 64)      | 24,832  |
| dense (Dense)                 | (None, 32)      | 2,080   |
| dropout (Dropout)             | (None, 32)      | 0       |
| dense_1 (Dense)               | (None, 25)      | 825     |

**Total params:** 1,043,213 (3.98 MB)

**Trainable params:** 347,737 (1.33 MB)

**Non-trainable params:** 0 (0.00 B)

**Optimizer params:** 695,476 (2.65 MB)

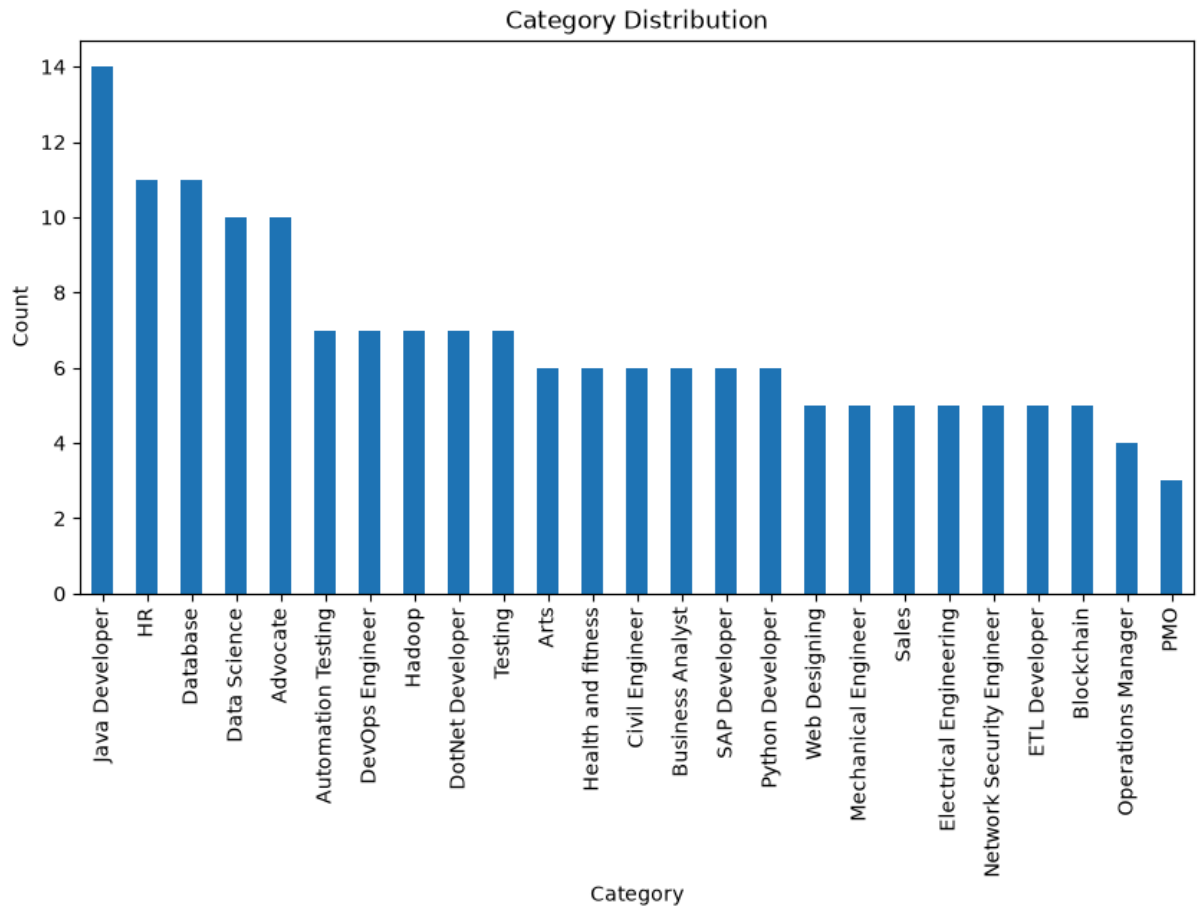
Explanation: Model Summary generated during fairness and explainability analysis.

### Category Counts

```
Category
Java Developer      14
HR                  11
Database            11
Data Science        10
Advocate            10
Automation Testing  7
DevOps Engineer     7
Hadoop              7
DotNet Developer    7
Testing             7
Arts                6
Health and fitness  6
Civil Engineer      6
Business Analyst    6
SAP Developer       6
Python Developer    6
Web Designing       5
Mechanical Engineer 5
Sales               5
Electrical Engineering 5
Network Security Engineer 5
ETL Developer       5
Blockchain          5
Operations Manager   4
PMO                 3
Name: count, dtype: int64
```

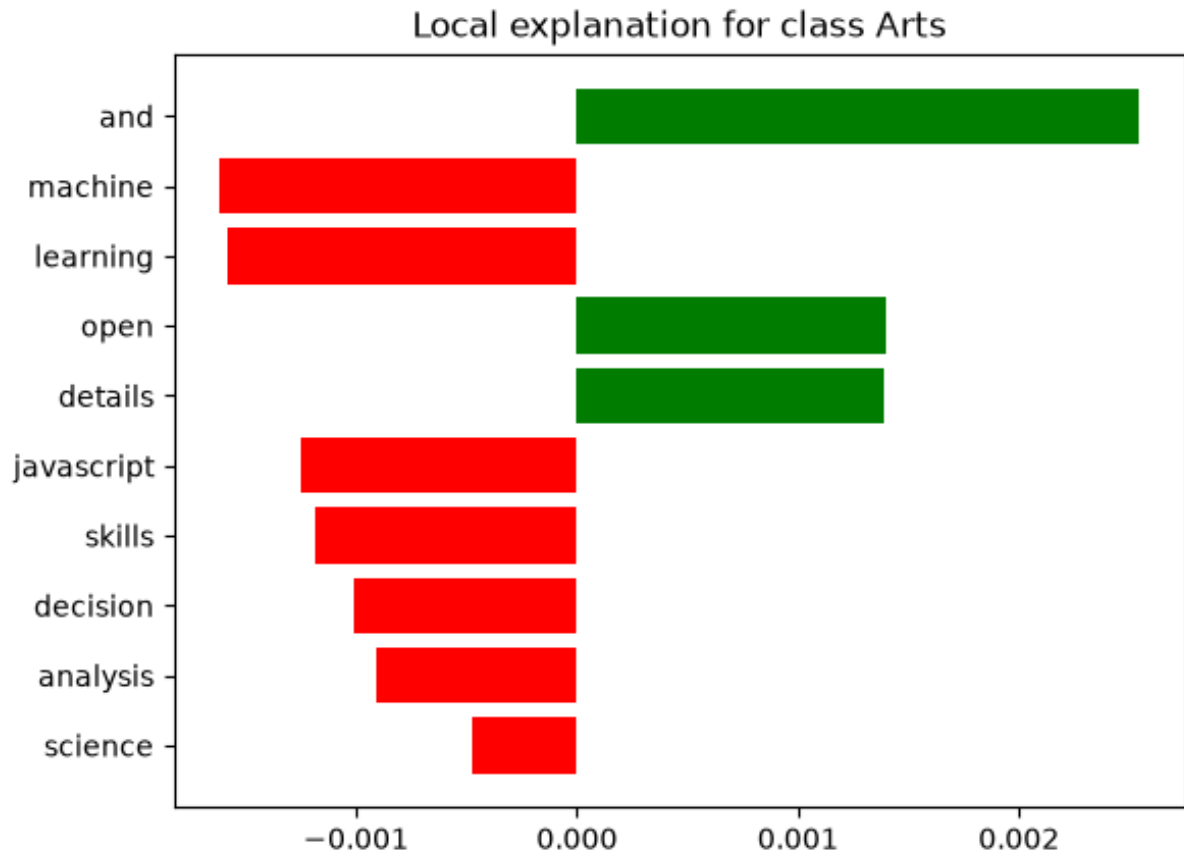
Explanation: Category Counts generated during fairness and explainability analysis.

### Category Distribution



Explanation: Category Distribution generated during fairness and explainability analysis.

LIME Explanation



Explanation: LIME Explanation generated during fairness and explainability analysis.

## 8. Bias Findings

The dataset is imbalanced. Categories like Java Developer and HR have more samples than PMO and Operations Manager. This may bias predictions.

## 9. Mitigation Recommendations

- Balance the dataset
- Increase category diversity
- Remove sensitive information
- Perform fairness checks regularly
- Collect more training samples

## 10. Conclusion

Responsible AI analysis was successfully performed. Explainability and bias detection techniques were applied to improve transparency and trustworthiness.